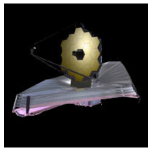

11218 - Tracing Reverse Shocked Ejecta and Dust Evolution in the Type Ia Supernova Remnant SNR 0509–67.5.

Cycle: 5, Proposal Category: GO

INVESTIGATORS

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OBSERVATIONS

Folder	*Observation*	*Label*	*Observing Template*	*Science Target*
Observation Folder				
	1	NIRspec IFU A	NIRSpec IFU Spectroscopy	(1) SNR0509-67.5-clumps1
	2	NIRSpec IFU Background	NIRSpec IFU Spectroscopy	(2) Background
	3	NIRSpec IFU B	NIRSpec IFU Spectroscopy	(3) SNR-0509-67.5-clumps2
	4	MIRI MRS A	MIRI Medium Resolution Spectroscopy	(4) SNR0509-67.5-MIRI-IFUA
	5	MIRI MRS background	MIRI Medium Resolution Spectroscopy	(5) MIRI-background
	6	MIRI MRS B	MIRI Medium Resolution Spectroscopy	(6) SNR0509-67.5-MIRI-IFUB

ABSTRACT

Type Ia supernovae play a central role in astrophysics, yet their explosion mechanisms, progenitor systems and how the ejecta evolve after the explosion, are open questions in stellar astrophysics that young remnants can uniquely answer. We propose a coordinated JWST NIRSpec IFU + MIRI MRS program to perform the first sensitive, spatially resolved infrared study of the young Type Ia supernova remnant SNR 0509–67.5 (LMC, age ~350 yr). NIRSpec IFU mapping will target faint, broad near infrared coronal lines from reverse shocked ejecta (e.g. [Ni xiii], [Si x], [Fe xiii], and successive S ionization stages) to trace the 3D distribution of iron-group and intermediate-mass elements. These maps will (i) test the double-detonation / sub-Chandrasekhar explosion interpretation inferred from recent optical tomography by comparing the spatial stratification of Si, S, Ca and Ni/Fe line ratios, and (ii) measure the size, velocity and ionization states of chemically distinct ejecta clumps to compare 3D hydrodynamic remnant models. MIRI MRS observations will deliver the first spatially resolved 5–28.8 m spectra of the eastern and western rims to search FeS, silicates, SiO features, constrain grain sizes and dust masses and distinguish the dust continuum between the forward and reverse shock. JWST's low background and IFU capabilities are essential to separate faint ejecta lines and continuum from nebular and background emission. Our results will help understand dust properties in a benchmark Type Ia remnant and provide a multi-wavelength, model-tested fingerprint of the explosion physics.

OBSERVING DESCRIPTION

We will use JWST NIRSpec IFU with the G140M/F100LP grating to obtain medium-resolution spectra across three adjacent pointing at SNR 0509–67.5, covering the clumpy ejecta region from the forward shock to the unshocked interior. These observations will target broad coronal and intermediate-mass element lines of iron, sulphur, nickel, and silicon originating from the reverse-shocked ejecta, enabling the first direct test of the predicted double-detonation origin of this Type Ia remnant. A matched dedicated NIRSpec background field and dithers per pointing are included to ensure accurate sky/background subtraction and improved spatial sampling. MIRI MRS IFU will provide full 4.9–28.8 μm spectral coverage (all four MRS channels over multiple exposures) of the limb-brightened eastern and western rim; locations evident to interact with a different CSM density, together with a dedicated background field for continuum and line subtraction. We also employ simultaneous MIRI imaging with F560W, F1000W and F1130W of the entire remnant and background. Together, these coordinated NIR–MIR datasets will deliver a comprehensive picture of the explosion physics, shock evolution, and dust processing in a Type Ia supernova remnant.

Fixed Targets

#	Name	Target Coordinates	Targ. Coord. Corrections	Miscellaneous
(1)	SNR0509-67.5-clumps1	RA: 05 09 33.1327 (77.3880529d) Dec: -67 31 18.25 (-67.52174d) Equinox: J2000	Epoch of Position: 2000	
Comments: Target RA and DEC has been chosen from archival MUSE observation of SNR 0509-67.5 *Category=ISM* *Description=[Supernova remnants]* *Extended=YES*				
(2)	Background	RA: 05 09 35.1150 (77.3963125d) Dec: -67 30 56.01 (-67.51556d) Equinox: J2000	Epoch of Position: 2000	
Comments: Target RA and DEC has been chosen from archival MUSE observation of SNR 0509-67.5 *Category=Calibration* *Description=[Telescope/sky background]*				
(3)	SNR-0509-67.5-clumps2	RA: 05 09 32.3410 (77.3847542d) Dec: -67 31 18.25 (-67.52174d) Equinox: J2000	Epoch of Position: 2000	
Comments: Target RA and DEC has been chosen from archival MUSE observation of SNR 0509-67.5 *Category=ISM* *Description=[Supernova remnants]* *Extended=YES*				
(4)	SNR0509-67.5-MIRI-IFUA	RA: 05 09 33.4170 (77.3892375d) Dec: -67 31 17.61 (-67.52156d) Equinox: J2000	Epoch of Position: 2000	
Comments: Target RA and DEC has been chosen from archival MUSE observation of SNR 0509-67.5 *Category=ISM* *Description=[Dust nebulae, Supernova remnants]*				
(5)	MIRI-background	RA: 05 09 28.1710 (77.3673792d) Dec: -67 31 44.01 (-67.52889d) Equinox: J2000	Epoch of Position: 2000	
Comments: Target RA and DEC has been chosen from archival MUSE observation of SNR 0509-67.5 *Category=Calibration* *Description=[Telescope/sky background]*				
(6)	SNR0509-67.5-MIRI-IFUB	RA: 05 09 29.0480 (77.3710333d) Dec: -67 31 21.62 (-67.52267d) Equinox: J2000	Epoch of Position: 2000	
Comments: Target RA and DEC has been chosen from archival MUSE observation of SNR 0509-67.5 *Category=ISM* *Description=[Supernova remnants]*				

Observation

Proposal 11218, Observation 1: NIRspec IFU A

Wed Apr 01 00:00:14 GMT 2026

Diagnostic Status: Warning

Observing Template: NIRSpec IFU Spectroscopy

Background Observations:[NIRSpec IFU Background (Obs 2)]

Diagnostics

(Visit 1:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.

Fixed Targets

#	Name	Target Coordinates	Targ. Coord. Corrections	Miscellaneous
(1)	SNR0509-67.5-clumps1	RA: 05 09 33.1327 (77.3880529d) Dec: -67 31 18.25 (-67.52174d) Equinox: J2000	Epoch of Position: 2000	

Comments: Target RA and DEC has been chosen from archival MUSE observation of SNR 0509-67.5
Category=ISM
Description=[Supernova remnants]
Extended=YES

Template

TA Method	HFF Readout Mode
NONE	false

Mosaic

Rows	Columns	Row Overlap %	Column Overlap %	Row shift (deg)	Column shift (deg)	Tile Order
2	1	10.0	10.0	0.0	0.0	DEFAULT

Dithers

#	Dither Type	Size	Starting Point	Number of Points	Points
1	4-POINT-DITHER				

Spectral Elements

#	Grating/Filter	Readout Pattern	Groups/Int	Integrations/Exp	Leakcal	Dither	Autocal	Total Dithers	Total Integrations	Total Exposure Time	Optional ETC ID
1	G140M/F100LP	NRSIRS2	18	5	false	true	NONE	4	20	26551.78	

Special Requirements	
	Aperture PA Range 83.97164917 to 93.97164917 Degrees (V3 305.0 to 315.0) Group Observations 1, 2, Non-interruptible

Observation

Proposal 11218, Observation 2: NIRSpec IFU Background

Wed Apr 01 00:00:14 GMT 2026

Diagnostic Status: Warning

Observing Template: NIRSpec IFU Spectroscopy

Background Observation For: [NIRspec IFU A (Obs 1)]

Diagnostics

(Visit 2:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.

Fixed Targets

#	Name	Target Coordinates	Targ. Coord. Corrections	Miscellaneous
(2)	Background	RA: 05 09 35.1150 (77.3963125d) Dec: -67 30 56.01 (-67.51556d) Equinox: J2000	Epoch of Position: 2000	

Comments: Target RA and DEC has been chosen from archival MUSE observation of SNR 0509-67.5
Category=Calibration
Description=[Telescope/sky background]

Template

TA Method	HFF Readout Mode
NONE	false

Dithers

#	Dither Type	Size	Starting Point	Number of Points	Points
1	4-POINT-DITHER				

Spectral Elements

#	Grating/Filter	Readout Pattern	Groups/Int	Integrations/Exp	Leakcal	Dither	Autocal	Total Dithers	Total Integrations	Total Exposure Time	Optional ETC ID
1	G140M/F100LP	NRSIRS2	18	5	false	true	NONE	4	20	26551.78	

Special Requirements

Group Observations 1, 2, Non-interruptible

Observation

Proposal 11218, Observation 3: NIRSpec IFU B

Wed Apr 01 00:00:14 GMT 2026

Diagnostic Status: Warning

Observing Template: NIRSpec IFU Spectroscopy

Diagnostics

(Visit 3:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.

Fixed Targets

#	Name	Target Coordinates	Targ. Coord. Corrections	Miscellaneous
(3)	SNR-0509-67.5-clumps2	RA: 05 09 32.3410 (77.3847542d) Dec: -67 31 18.25 (-67.52174d) Equinox: J2000	Epoch of Position: 2000	

Comments: Target RA and DEC has been chosen from archival MUSE observation of SNR 0509-67.5
Category=ISM
Description=[Supernova remnants]
Extended=YES

Template

TA Method	HFF Readout Mode
NONE	false

Dithers

#	Dither Type	Size	Starting Point	Number of Points	Points
1	4-POINT-DITHER				

Spectral Elements

#	Grating/Filter	Readout Pattern	Groups/Int	Integrations/Exp	Leakcal	Dither	Autocal	Total Dithers	Total Integrations	Total Exposure Time	Optional ETC ID
1	G140M/F100LP	NRSIRS2	18	5	false	true	NONE	4	20	26551.78	

Special Requirements

Aperture PA Range 83.97164917 to 93.97164917 Degrees (V3 305.0 to 315.0)

Observation

Proposal 11218, Observation 4: MIRI MRS A

Wed Apr 01 00:00:14 GMT 2026

Diagnostic Status: Warning

Observing Template: MIRI Medium Resolution Spectroscopy

Background Observations:[MIRI MRS background (Obs 5)]

Diagnostics

(Visit 4:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.

Fixed Targets

#	Name	Target Coordinates	Targ. Coord. Corrections	Miscellaneous
(4)	SNR0509-67.5-MIRI-IFUA	RA: 05 09 33.4170 (77.3892375d) Dec: -67 31 17.61 (-67.52156d) Equinox: J2000	Epoch of Position: 2000	

Comments: Target RA and DEC has been chosen from archival MUSE observation of SNR 0509-67.5
Category=ISM
Description=[Dust nebulae, Supernova remnants]

Acquisition

#	Target
1	NONE

Template

AcqFilter	Primary Channel	Simultaneous Imaging	Imager Subarray	Grating Wheel Direction
F560W	All MRS	YES	FULL	Allow Auto Reorder

Dithers

#	Dither Type	Optimized For	Direction
1	4-Point	POINT SOURCE	NEGATIVE

Spectral Elements

#	Wavelength Range	Detector	Filter	Readout Pattern	Groups/Int	Integrations/Exp	Exposures/Dith	Dither	Total Dithers	Total Integrations	Total Exposure Time	Optional ETC ID
1		IMAGER	F560W	FASTR1	15	1	1	Dither 1	4	4	166.502	
1	SHORT(A)	MRSLONG		FASTR1	50	1	1	Dither 1	4	4	555.008	
1	SHORT(A)	MRSSHORT		FASTR1	50	1	1	Dither 1	4	4	555.008	
2		IMAGER	F1000W	FASTR1	15	1	1	Dither 1	4	4	166.502	
2	MEDIUM(B)	MRSLONG		FASTR1	50	1	1	Dither 1	4	4	555.008	
2	MEDIUM(B)	MRSSHORT		FASTR1	50	1	1	Dither 1	4	4	555.008	
3		IMAGER	F1130W	FASTR1	15	1	1	Dither 1	4	4	166.502	
3	LONG(C)	MRSLONG		FASTR1	50	1	1	Dither 1	4	4	555.008	
3	LONG(C)	MRSSHORT		FASTR1	50	1	1	Dither 1	4	4	555.008	

Special Requirements	
	Offset -66.32464021961947 arcsec, -58.0465404357874 arcsec Group Observations 4, 5, 6, Non-interruptible

Observation

Proposal 11218, Observation 5: MIRI MRS background

Wed Apr 01 00:00:14 GMT 2026

Diagnostic Status: Warning

Observing Template: MIRI Medium Resolution Spectroscopy

Background Observation For: [MIRI MRS A (Obs 4), MIRI MRS B (Obs 6)]

Diagnostics

(Visit 5:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.

Fixed Targets

#	Name	Target Coordinates	Targ. Coord. Corrections	Miscellaneous
(5)	MIRI-background	RA: 05 09 28.1710 (77.3673792d) Dec: -67 31 44.01 (-67.52889d) Equinox: J2000	Epoch of Position: 2000	

Comments: Target RA and DEC has been chosen from archival MUSE observation of SNR 0509-67.5
Category=Calibration
Description=[Telescope/sky background]

Acquisition

#	Target
1	NONE

Template

AcqFilter	Primary Channel	Simultaneous Imaging	Imager Subarray	Grating Wheel Direction
	All MRS	YES	FULL	Allow Auto Reorder

Dithers

#	Dither Type	Optimized For	Direction
1	4-Point	POINT SOURCE	NEGATIVE

Spectral Elements

#	Wavelength Range	Detector	Filter	Readout Pattern	Groups/Int	Integrations/Exp	Exposures/Dith	Dither	Total Dithers	Total Integrations	Total Exposure Time	Optional ETC ID
1		IMAGER	F560W	FASTR1	15	1	1	Dither 1	4	4	166.502	
1	SHORT(A)	MRSLONG		FASTR1	50	1	1	Dither 1	4	4	555.008	
1	SHORT(A)	MRSSHORT		FASTR1	50	1	1	Dither 1	4	4	555.008	
2		IMAGER	F1000W	FASTR1	15	1	1	Dither 1	4	4	166.502	
2	MEDIUM(B)	MRSLONG		FASTR1	50	1	1	Dither 1	4	4	555.008	
2	MEDIUM(B)	MRSSHORT		FASTR1	50	1	1	Dither 1	4	4	555.008	
3		IMAGER	F1130W	FASTR1	15	1	1	Dither 1	4	4	166.502	
3	LONG(C)	MRSLONG		FASTR1	50	1	1	Dither 1	4	4	555.008	
3	LONG(C)	MRSSHORT		FASTR1	50	1	1	Dither 1	4	4	555.008	

Special Requirements	
	Offset -48.9907764207058 arcsec, 35.49546256533064 arcsec Group Observations 4, 5, 6, Non-interruptible

Observation

Proposal 11218, Observation 6: MIRI MRS B

Wed Apr 01 00:00:14 GMT 2026

Diagnostic Status: Warning

Observing Template: MIRI Medium Resolution Spectroscopy

Background Observations:[MIRI MRS background (Obs 5)]

Diagnostics

(Visit 6:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.

Fixed Targets

#	Name	Target Coordinates	Targ. Coord. Corrections	Miscellaneous
(6)	SNR0509-67.5-MIRI-IFUB	RA: 05 09 29.0480 (77.3710333d) Dec: -67 31 21.62 (-67.52267d) Equinox: J2000	Epoch of Position: 2000	

Comments: Target RA and DEC has been chosen from archival MUSE observation of SNR 0509-67.5
Category=ISM
Description=[Supernova remnants]

Acquisition

#	Target
1	NONE

Template

AcqFilter	Primary Channel	Simultaneous Imaging	Imager Subarray	Grating Wheel Direction
	All MRS	YES	FULL	Allow Auto Reorder

Dithers

#	Dither Type	Optimized For	Direction
1	4-Point	POINT SOURCE	NEGATIVE

Spectral Elements

#	Wavelength Range	Detector	Filter	Readout Pattern	Groups/Int	Integrations/Exp	Exposures/Dith	Dither	Total Dithers	Total Integrations	Total Exposure Time	Optional ETC ID
1		IMAGER	F560W	FASTR1	15	1	1	Dither 1	4	4	166.502	
1	SHORT(A)	MRSLONG		FASTR1	50	1	1	Dither 1	4	4	555.008	
1	SHORT(A)	MRSSHORT		FASTR1	50	1	1	Dither 1	4	4	555.008	
2		IMAGER	F1000W	FASTR1	15	1	1	Dither 1	4	4	166.502	
2	MEDIUM(B)	MRSLONG		FASTR1	50	1	1	Dither 1	4	4	555.008	
2	MEDIUM(B)	MRSSHORT		FASTR1	50	1	1	Dither 1	4	4	555.008	
3		IMAGER	F1130W	FASTR1	15	1	1	Dither 1	4	4	166.502	
3	LONG(C)	MRSLONG		FASTR1	50	1	1	Dither 1	4	4	555.008	
3	LONG(C)	MRSSHORT		FASTR1	50	1	1	Dither 1	4	4	555.008	

Special Requirements	
	Offset -37.7671134287436 84 arcsec, -57.02175814475368 arcsec Group Observations 4, 5, 6, Non-interruptible